

DSA 210 Spring

**Analyzing the Relationship Between Screen Time and Sleep
Patterns Using Personal Behavioral Data**

Final Report

İrfan Efe İnal

32381

1. Motivation

The increasing use of smartphones and digital devices has made screen exposure a significant part of daily life. Many studies and public discussions suggest that excessive screen usage may negatively affect sleep quality and sleep duration. Since smartphones have become deeply integrated into everyday routines, understanding whether screen time is associated with sleep behavior is an interesting and relevant problem from a data science perspective.

The motivation behind this project was to investigate whether there exists a measurable relationship between daily screen time and sleep duration using real personal behavioral data. Instead of relying on public datasets, this project uses personal device usage records to provide an individualized perspective on digital habits and sleep behavior. The project aims not only to identify possible patterns but also to apply the complete data science pipeline, including data collection, preprocessing, statistical analysis, visualization, and machine learning.

2. Data Source

The dataset used in this project was collected from personal smartphone usage records. Daily screen time values were obtained from the iPhone Screen Time feature, which reports total device usage duration for each day. Direct historical sleep duration information was not available in an easily exportable format. Therefore, sleep duration was estimated using a proxy based on periods of phone inactivity. Since the first activity after waking up and the last activity before sleeping generally involved checking the phone, the longest overnight period without phone activity was used as an approximation of sleep duration. The dataset consists of daily observations collected over approximately one month, from April 1st to May 5th, totaling 35 observations.

The main variables included:

- Date
- Daily screen time
- Estimated sleep duration

Additional features were created during preprocessing:

- Screen time in minutes
- Weekday information
- Weekend indicator variable
- Binary sleep quality indicator (“good sleep” if sleep duration ≥ 7 hours)

Although the dataset is relatively small, it contains realistic behavioral variability and represents authentic personal usage patterns.

3. Data Analysis

The project followed multiple stages of the data science workflow.

3.1 Data Preprocessing

Raw screen time values were initially stored in hour-minute format (e.g., “5h 53m”). To allow quantitative analysis, these values were converted into total minute Sleep duration values were similarly transformed into numerical hour values. Additional feature engineering was applied:

- Weekday extraction
- Weekend identification

- Sleep quality categorization

Missing values and invalid observations were removed before analysis.

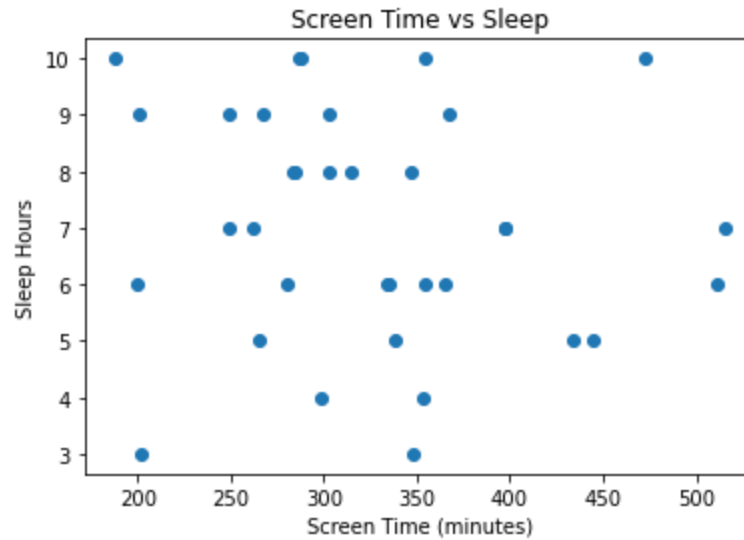
During the implementation process, AI-based tools were used for debugging Python code, improving code organization, and refining language. Suggestions generated by these tools were reviewed and validated before use, while all decisions regarding data collection, preprocessing, analysis methods, and interpretation of results were made by the author.

3.2 Exploratory Data Analysis (EDA)

Several visualizations were generated to understand the characteristics of the dataset. A scatter plot was used to visualize the relationship between screen time and sleep duration. Time series graphs were also created to observe how screen time and sleep duration changed over time.

Descriptive statistics showed:

Metric	Screen Time (minutes)	Sleep Duration (hours)
Count	35	35
Mean	325.46	7.03
Standard deviation	82.94	2.05
Minimum	188	3
Maximum	515	10



(Figure 1)

Figure 1 illustrates the relationship between screen time and sleep duration. A weak negative trend can be observed, indicating that increased screen time may be associated with reduced sleep duration. However, the data points are relatively dispersed, suggesting that the relationship is not particularly strong.

The correlation coefficient between screen time and sleep duration was: $r = -0.129$

This indicates a weak negative relationship between the two variables.

3.3 Hypothesis Testing

To statistically investigate whether screen time influences sleep duration, a two-sample t-test was performed.

Null hypothesis:

H_0 : There is no significant difference in sleep duration between high screen time days and low screen time days.

Alternative hypothesis:

H_1 : There is a significant difference in sleep duration between high screen time days and low screen time days.

Days were divided into two groups using the average screen time as the threshold.

Results:

- $p\text{-value} = 0.1185$

Since the p-value is greater than 0.05, the null hypothesis cannot be rejected.

Therefore, the observed difference between high and low screen-time days is not statistically significant.

3.4 Machine Learning Model

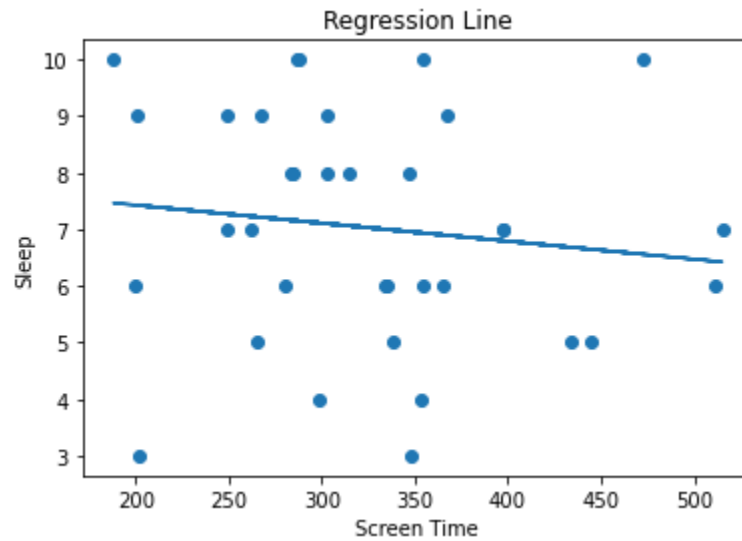
A linear regression model was implemented to predict sleep duration using:

- Screen time
- Weekend information

The dataset was divided into training and testing sets using a 70%-30% split.

Model performance was evaluated using:

- Mean Absolute Error (MAE)
- R^2 score



(Figure 2)

Figure 2 shows the linear regression model fitted to the observed data. The regression line has a slightly negative slope, suggesting that higher screen time tends to correspond with lower sleep duration. However, the weak slope and dispersed observations indicate limited predictive power.

Results:

- MAE = 1.95 hours
- $R^2 = -0.0119$

The model coefficient for screen time was negative:

- Screen time coefficient = -0.00455

This suggests that higher screen time tends to reduce sleep duration. However, the predictive performance of the model was weak.

4. Findings

The results suggest that there exists a weak negative relationship between screen time and sleep duration. The scatter plot and regression analysis indicate that sleep duration tends to decrease as screen time increases. However, the relationship is relatively weak and the data points exhibit considerable variability. The hypothesis testing results further support this observation. Although differences between high and low screen time groups were observed, these differences were not statistically significant. The machine learning model also showed limited predictive power. The negative R^2 value indicates that screen time alone is insufficient for accurately predicting sleep behavior. This suggests that sleep patterns are likely influenced by multiple factors beyond screen exposure. Although the relationship observed in this study was weak, the findings demonstrate how personal behavioral data can be analyzed using statistical and machine learning methods to uncover potential patterns.

Key findings:

- Average screen time: 325.46 minutes/day
- Average sleep duration: 7.03 hours/day
- Correlation coefficient: -0.129
- p-value: 0.118
- R^2 score: -0.0119
- A weak negative relationship was observed.

5. Limitations and Future Work

Several limitations exist within this study. First, the dataset consists of only 35 observations, which limits statistical power and model performance. Second, sleep duration was approximated using phone inactivity intervals rather than directly measured sleep data. While this approach provides a reasonable estimate, it may introduce measurement errors. Third, the dataset contains only personal behavioral data from a single individual. Consequently, the findings cannot be generalized to larger populations.

Future work may include:

- Collecting a larger dataset over multiple months
- Using Apple Watch sleep measurements instead of approximations
- Including additional features such as:
 - physical activity
 - stress levels
 - caffeine consumption
 - academic workload
 - mood indicators

More advanced machine learning models such as Random Forest or Classification methods may also improve predictive performance.

6. Conclusion

This project implemented a complete data science workflow to investigate the relationship between screen time and sleep behavior using personal behavioral data. The findings indicate a weak negative association between screen time and sleep duration, but the results were not statistically strong enough to establish a definitive conclusion. Despite the limitations, the project successfully demonstrated the process of collecting data, preprocessing it, applying statistical techniques, building machine learning models, and interpreting findings. The results suggest that screen time alone does not fully explain sleep behavior and that additional factors should be considered in future studies.